

Permutation Prime Sums

Time limit: 1.00 s

Memory limit: 512 MB

Given n , create two permutations a and b of size n such that $a_i + b_i$ is prime for $i = 1, 2, \dots, n$.

Input

The only line has an integer n .

Output

Print two permutations. You can print any valid solution. If there are no solutions, print IMPOSSIBLE.

Constraints

- $1 \leq n \leq 10^5$

Example

Input	Output
5	2 1 3 5 4 5 1 4 2 3

Explanation: The sums are $2 + 5 = 7$, $1 + 1 = 2$, $3 + 4 = 7$, $5 + 2 = 7$ and $4 + 3 = 7$ which all are primes.